

Transcendental Observation Grids, Bond Pricing, and the Reconstruction of Stochastic Discount Factors

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Abstract

This paper develops a framework for reconstructing discount functions and stochastic discount factors (SDFs) from observations of a bond at a collection of transcendental observation times generated by the constants π , e , and ϕ . We derive the associated observation grid, prove basic identification results, construct parametric and nonparametric estimators of the pricing kernel, and discuss implications for economics, finance, statistics, machine learning, geopolitics, warfare economics, and epistemology. The resulting framework may be interpreted as a sparse quadrature system for extracting latent intertemporal valuation structures from limited observations.

The paper ends with “The End”

1 Introduction

Asset pricing theory is fundamentally concerned with the valuation of future cash flows.

For an asset with future payoff stream X ,

$$P_t = \mathbb{E}_t[M_{t,T}X_T],$$

where $M_{t,T}$ denotes the stochastic discount factor.

The central problem is inversion:

$$\text{Observed Prices} \implies \text{Discount Function} \implies \text{SDF}.$$

This paper investigates a special observation scheme generated by transcendental constants:

$$\pi, \quad e, \quad \phi.$$

These constants appear throughout mathematics, physics, information theory, optimization, and biological growth. We show that they generate a useful observation grid for extracting discount information from bond prices.

2 The Transcendental Observation Grid

Let

$$t_1 < t_2.$$

Define

$$\Delta t = t_2 - t_1.$$

For any constant c define

$$\frac{t_c - t_1}{t_2 - t_1} = \frac{1}{c}.$$

Solving yields

$$t_c = t_1 + \frac{\Delta t}{c}.$$

We consider the constants

$$\pi, \quad \pi - 1, \quad e, \quad e - 1, \quad \phi, \quad \phi - 1.$$

The associated observation times are

$$\begin{aligned} t_\pi &= t_1 + \frac{\Delta t}{\pi}, \\ t_{\pi-1} &= t_1 + \frac{\Delta t}{\pi - 1}, \\ t_e &= t_1 + \frac{\Delta t}{e}, \\ t_{e-1} &= t_1 + \frac{\Delta t}{e - 1}, \\ t_\phi &= t_1 + \frac{\Delta t}{\phi}, \\ t_{\phi-1} &= t_1 + \phi \Delta t. \end{aligned}$$

Introduce normalized time

$$u(t) = \frac{t - t_1}{t_2 - t_1}.$$

The grid becomes

$$\mathcal{U} = \left\{ 0, \frac{1}{\pi}, \frac{1}{e}, \frac{1}{\pi - 1}, \frac{1}{e - 1}, \frac{1}{\phi}, 1, \phi \right\}.$$

3 Numerical Structure

Node	Value
0	0.000000
$1/\pi$	0.318310
$1/e$	0.367879
$1/(\pi - 1)$	0.466942
$1/(e - 1)$	0.581977
$1/\phi$	0.618034
1	1.000000
ϕ	1.618034

Table 1: Transcendental observation grid.

Observe the concentration of points in the interval

$$0.3 < u < 0.62.$$

This creates a non-uniform sampling geometry.

4 Bond Pricing

Consider a bond with deterministic cash flows

$$C_1, \dots, C_n.$$

The value at time t is

$$A(t) = \sum_{j:T_j > t} D(t, T_j) C_j.$$

For a zero-coupon bond,

$$A(T) = F.$$

Hence

$$A(t) = F D(t, T).$$

The discount factor is therefore observed directly:

$$D(t, T) = \frac{A(t)}{F}.$$

5 Exact Recovery of the Discount Curve

Suppose observations are available at all grid points

$$u_1, \dots, u_8.$$

Define

$$D_i = \frac{A(u_i)}{F}.$$

Then

$$\mathcal{D} = \{D_1, \dots, D_8\}$$

constitutes eight exact observations of the discount curve.

Proposition 1. *For a deterministic zero-coupon bond, the discount function is pointwise identified at all observation nodes.*

Proof. Since

$$A(t) = F D(t, T),$$

and F is known,

$$D(t, T) = A(t)/F.$$

Therefore every observed price yields the corresponding discount factor exactly. $\square$

6 The Stochastic Discount Factor

The fundamental pricing equation is

$$A(t) = F \mathbb{E}_t[M(t, T)].$$

Thus

$$\mathbb{E}_t[M(t, T)] = \frac{A(t)}{F}.$$

Consequently,

$$\boxed{\mathbb{E}_t[M(t, T)] = D(t, T)}$$

for deterministic payoff bonds.

7 Parametric Estimation

Assume

$$D(u) = e^{-ru}.$$

Taking logarithms gives

$$\ln D(u) = -ru.$$

Given observations

$$(u_i, D_i),$$

the least-squares estimator is

$$\hat{r} = -\frac{\sum_i u_i \ln D_i}{\sum_i u_i^2}.$$

The estimated pricing kernel becomes

$$\widehat{M}(u) = e^{-\hat{r}u}.$$

8 Nonparametric Reconstruction

Let

$$B_k(u)$$

denote basis functions.

Define

$$\ln D(u) = \sum_{k=1}^8 \alpha_k B_k(u).$$

The coefficients satisfy

$$\ln D_i = \sum_{k=1}^8 \alpha_k B_k(u_i).$$

Hence

$$M(u) = \exp \left(\sum_{k=1}^8 \alpha_k B_k(u) \right).$$

9 A Transcendental-Factor Model

Define factors

$$f_\pi(u) = \frac{u}{1/\pi},$$

$$f_e(u) = \frac{u}{1/e},$$

$$f_\phi(u) = \frac{u}{1/\phi}.$$

Consider

$$M(u) = \exp(-\beta_\pi f_\pi(u) - \beta_e f_e(u) - \beta_\phi f_\phi(u)).$$

The coefficients

$$(\beta_\pi, \beta_e, \beta_\phi)$$

may be estimated from the observed discount factors.

10 Statistical Interpretation

The observation grid functions as a sparse quadrature design.

The estimation problem becomes

$$\min_{\theta} \sum_{i=1}^8 \left(D_i - \hat{D}(u_i; \theta) \right)^2.$$

The Fisher information matrix is

$$I(\theta) = \sum_i \left(\frac{\partial D_i}{\partial \theta} \right) \left(\frac{\partial D_i}{\partial \theta} \right)^T.$$

Grid quality may be measured through

$$\det I(\theta).$$

11 Machine Learning Interpretation

The discount function may be viewed as an unknown target function

$$D : \mathbb{R} \rightarrow \mathbb{R}.$$

The observation nodes become training data:

$$\{(u_i, D_i)\}_{i=1}^8.$$

A neural approximation is

$$D(u) \approx N_\theta(u).$$

Training solves

$$\min_{\theta} \sum_i (N_\theta(u_i) - D_i)^2.$$

The transcendental grid therefore acts as a feature-engineered sampling scheme.

12 Economic Interpretation

The SDF represents intertemporal scarcity.

Large discount factors imply

Future Consumption Highly Valued.

Small discount factors imply

Present Consumption Highly Valued.

The grid extracts information about the shape of intertemporal preferences.

13 Political Economy and Geopolitics

States, empires, and military organizations implicitly employ discount functions.

For a strategic actor,

$$V = \sum_t M_t X_t.$$

Short-horizon powers exhibit steep discounting.

Civilizational actors exhibit flatter discounting.

Examples include:

- infrastructure investment,
- military modernization,
- demographic planning,
- sovereign debt management,
- technological competition.

Thus the SDF serves as a quantitative representation of strategic patience.

14 Warfare Economics

Military procurement can be written as

$$V_{\text{war}} = \sum_t M_t B_t.$$

Here B_t denotes expected strategic benefit.

Estimating the military discount function allows one to infer:

1. tolerance for delayed gains,
2. expected conflict duration,
3. deterrence credibility,
4. escalation incentives.

The same mathematical structure underlying bond valuation appears in defense economics.

15 Philosophical Interpretation

The SDF is fundamentally a valuation operator.

It transforms future possibilities into present realities.

One may therefore interpret

$$M(t, T)$$

as a measure of temporal belief.

Questions of economics become questions of epistemology:

How much is the future worth?

and

How certain are we that it will arrive?

16 Geometry of the Grid



17 Main Identification Theorem

Theorem 1. *Let a zero-coupon bond possess known face value F and maturity date T . Suppose prices are observed at the transcendental grid nodes*

$$\mathcal{U} = \left\{ 0, \frac{1}{\pi}, \frac{1}{e}, \frac{1}{\pi-1}, \frac{1}{e-1}, \frac{1}{\phi}, 1, \phi \right\}.$$

Then the discount function is exactly identified at all grid points by

$$D_i = \frac{A_i}{F}.$$

Furthermore,

$$\mathbb{E}[M_i] = D_i.$$

Proof. Immediate from

$$A_i = F D_i.$$

Dividing by F yields exact recovery of the discount factor.
Since

$$A_i = F \mathbb{E}[M_i],$$

the expectation of the SDF equals the discount factor. □

18 Conclusion

The transcendental observation grid generated by π , e , and ϕ yields a structured sparse sampling design for discount-factor reconstruction. For deterministic-payoff bonds, the discount curve is directly recoverable from prices. The resulting framework admits interpretations in finance, economics, statistics, machine learning, strategic studies, geopolitics, warfare economics, and philosophy. The stochastic discount factor emerges as a universal valuation operator linking future states of the world to present decisions.

References

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Glossary

SDF

Stochastic Discount Factor.

Discount Factor

Present value of one unit of future payoff.

Pricing Kernel

Alternative name for the SDF.

Zero-Coupon Bond

Bond paying a single maturity value.

Transcendental Grid

Observation scheme generated by π , e , and ϕ .

Quadrature

Numerical approximation of integrals using sampled points.

Strategic Patience

Implicit discount rate of a political or military actor.

Intertemporal Valuation

Assignment of value across different dates.

The End